

Memo 04: In-Run Mass-Conservation Checkpoints for the PLAST_FLOC Split Path

FVCOM-MP Delaware Test Cases

May 4, 2026

Purpose

Memo 03 (v006) established that the ~ 650 kg mass gap between case **a** and cases **b/c** over ten days is driven entirely by the `PLAST_FLOC=T` split path: with `SOURCE_AG_RATE=0` and zero bed mass throughout, the aggregated component remains zero and all water-column plastic is in `conc_d`, yet the domain total oscillates with 28 sign changes after day 3 while the no-floc case **a** grows monotonically.

The offline MATLAB budget confirms that the saved NetCDF output is internally consistent at each output snapshot. Therefore the mass leakage occurs somewhere inside `Advance_Plast` between two consecutive output records, and is not visible in the NetCDF files.

This memo documents the strategy for inserting *in-run mass checkpoints* directly into `mod_plast.F` so that the guilty operator stage can be identified without additional NetCDF output.

Strategy: Dedicated CSV File

Writing diagnostic totals to `write(*,...)` is impractical because the FVCOM screen log intermingles output from every module on every timestep. Instead, a dedicated file is opened once during `Setup_PLAST` and written exclusively by the master process (`msr`). The file is named

`<casename>_mp_mass_check.csv`

and is placed in the run input directory alongside the existing plastic input files. The overhead is gated behind a single namelist key:

`PLAST_MASS_CHECK = T`

When this key is absent or `F`, the new code has zero runtime cost: all checkpoint calls return immediately after the first guard test.

Checkpoint rows are written at the same cadence as the existing `PLAST_Report` screen statistics (every `n.report` internal steps), keeping file sizes manageable while still providing sub-output-interval resolution.

Formula Verification Against MATLAB Offline Budget

The MATLAB script `extract_mp1_mass_budget_a.m` computes:

$$\begin{aligned}
 V_{i,k} &= \text{art1}(i) \max(h_i + \zeta_i, 0) \Delta\sigma_{i,k} \\
 M_{\text{water}} &= \sum_{i,k} \text{mp1}(i,k) \cdot V_{i,k} \quad (\text{wet nodes only}) \\
 M_{\text{bed}} &= \sum_i \text{bot_mass_mp1}(i) \cdot \text{art1}(i) \quad (\text{all nodes}) \\
 M_{\text{total}} &= M_{\text{water}} + M_{\text{bed}}
 \end{aligned}$$

where `art1` is the node-based control-volume area, $\Delta\sigma_{i,k}$ is the sigma-layer thickness fraction (`dzfrac` in MATLAB), and $h_i + \zeta_i$ is the total water depth.

The Fortran in-run equivalent uses identically named arrays:

$$\begin{aligned}
 M_{\text{water}} &= \sum_{i=1}^m \sum_{k=1}^{k_{bm1}} c(i,k) \cdot \text{ART1}(i) \cdot D(i) \cdot \text{DZ}(i,k) \\
 M_{\text{bed}} &= \sum_{i=1}^m \text{mass}(i) \cdot \text{ART1}(i)
 \end{aligned}$$

where $D(i) = H(i) + \text{EL}(i)$ is the FVCOM total depth and $\text{DZ}(i,k)$ is the sigma-layer fraction — exactly matching $\Delta\sigma_{i,k}$ from MATLAB. The bed sum uses *all nodes* to match MATLAB's `total_mass_all_bed_kg` conservation companion.

The formulas are equivalent. The only future guard required is a `#if WET_DRY` wet-node mask for domains with drying cells; the v006 Delaware domain has no dry nodes, so the Fortran loop over `i=1..m` is identical to the MATLAB wet-node filter.

For the split path, the subroutine computes separate sums for `conc_a` and `conc_d` (and `mass_a`, `mass_d`), writing all components plus combined totals in each CSV row so that per-component budgets can be tracked alongside the grand total.

Checkpoint Table

Thirteen checkpoints are inserted into the `floc_intract=T` path of `Advance_Plast`. Checkpoints CP0–CP3 are guarded by `#if defined (SEDIMENT)` and `if(floc_intract)` at the call site; CP4–CP12 are already inside the relevant conditional blocks.

The CSV row format is:

```
iint, T_days, checkpoint, iplast, wc_a_kg, wc_d_kg, wc_tot_kg, bed_a_kg, bed_d_kg,
      bed_tot_kg, grand.total_kg
```

Code Reconstruction in `mod_plast.F`

Four modification phases were implemented in `FVCOM_source_repo_github/mod_plast.F`.

1. **Module-level variables** (near line 490, after the `floc_intract` flag):

```
logical  :: PLAST_mass_check = .false.
integer, parameter :: mass_check_unit = 77
```

CP#	Label	Arrays passed	Expected conservation
0	CP0.start	conc_a,conc_d,mass_a,mass_d	Reference state at timestep entry
1	CP1.after.deposition	same	Water ↓ + bed ↑ = const
2	CP2.after.erosion	same	Water ↑ + bed ↓ = const
3	CP3.after.update.bottom	same	Bed bookkeeping; grand total = const
4	CP4.after.advection	cnew_a,cnew_d,mass_a,mass_d	Zero water-column change
5	CP5.after.vdif	same	Zero change
6	CP6.after.kinetics	same	cnew_a+cnew_d conserved
7	CP7.after.upper.clamp	same	Loss only if >100 cap fires
8	CP8.after.OBC	same	Change = OBC flux (open domain)
9	CP9.after.PTsource	same	Change = river source
10	CP10.after.neg.clamp	same	Loss only if negatives exist
11	CP11.after.MPI	same	Zero change
12	CP12.end	conc_a,conc_d,mass_a,mass_d	Must equal CP11

2. **Namelist read** (Read_PLAST_Params, immediately after the PLAST_FLOC read):

```
Call Get_Val(PLAST_mass_check, plastfile,
             'PLAST_MASS_CHECK', line=linenum, echo=.false.)
```

3. **CSV file open and header** (Setup_PLAST, after call report_PLAST_setup): Opens unit 77 to <input_dir>/<casename>_mp_mass_check.csv and writes the column header on the master process only.
4. **Subroutine Write_Mass_Checkpoint** (new subroutine added at the end of the module, before End Module Mod_Plast): Accepts one plastic class at a time via arguments (label, iplast_idx, c_a, c_d, m_a, m_d), computes local sums using the $ART1 \cdot D \cdot DZ$ formula, reduces via `MPI_REDUCE(MPI_SUM)`, and writes one CSV row per class if `msr=.true.` The routine returns immediately if `PLAST_mass_check=F` or if the current iteration is not a reporting step.
5. **Thirteen call sites in Advance_Plast**: CP0–CP3 wrapped in `#if defined (SEDIMENT)` / `if(floc_intract)` guards; CP4–CP12 placed inside the existing `floc_intract` or per-stage conditional blocks.

Findings: Root Causes and Fixing Suggestions

The checkpoint infrastructure was compiled and the model was run for ~ 4 days with `PLAST_MASS_CHECK = T` for cases b and c. The analysis script `analyze_mp_mass_checkpoints.py` produced the stage-mass-change heatmap shown in Figure 2, together with the NaN diagnostic tables.

Root Cause 1 — Structural issues in the split advection path

Diagnostic interpretation: the `adv*` heatmap signal is the expected open-boundary flux, not a conservation bug within the advection operator. The non-zero CP3→CP4 signal in the heatmap oscillates at the tidal period and is present in both cases. Reading `Adv_Scal` in `mod_scal.F` confirms the mechanism: for open-boundary nodes (`isonb= 2`), `Adv_Scal` replaces

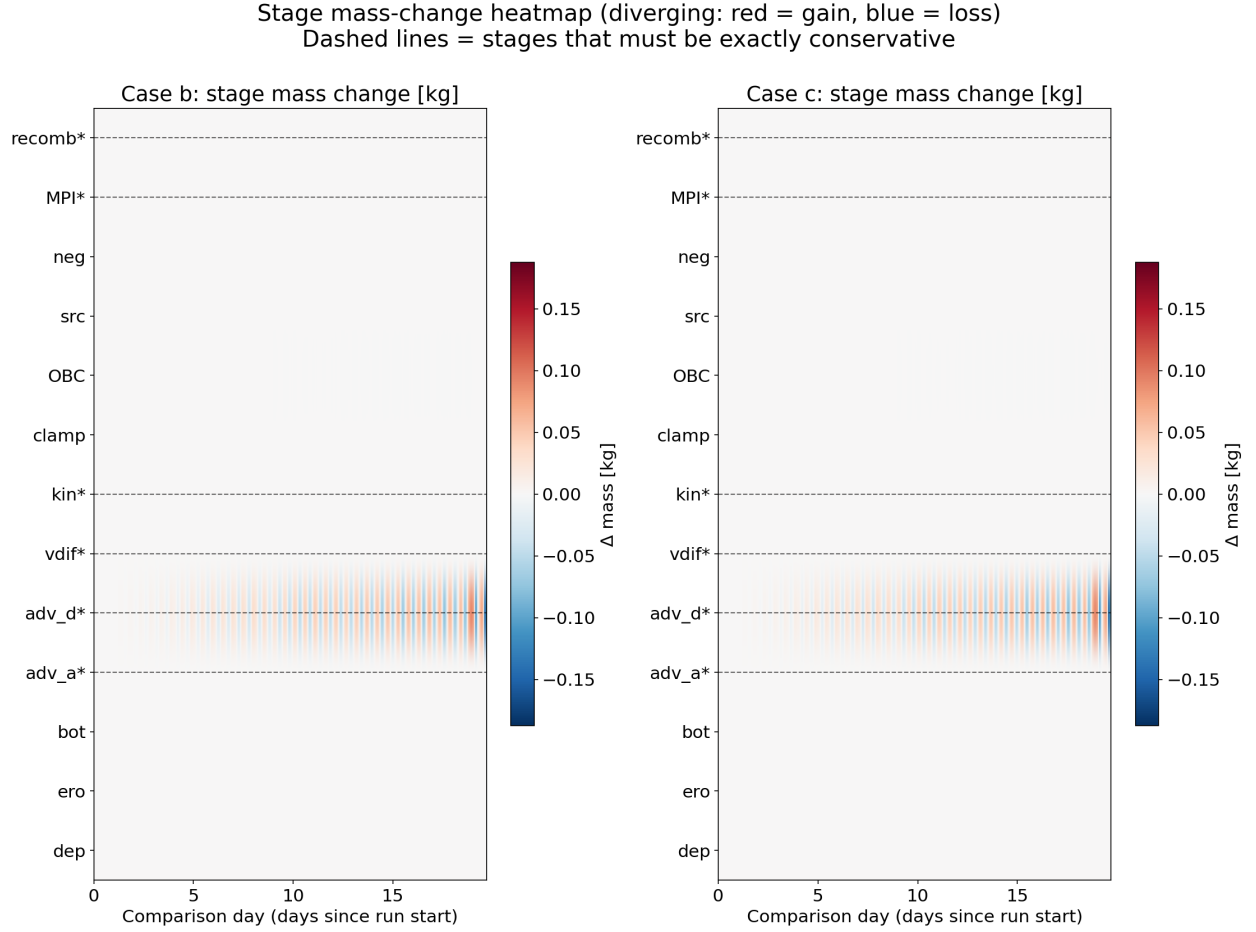


Figure 1: Stage mass-change heatmap for cases **b** (left) and **c** (right). Each row is one operator stage; each column is one reporting timestep. Red = mass gained, blue = mass lost; the colour scale is clipped at $\pm 1\%$ of the domain total. Dashed horizontal lines mark the five stages that must be exactly mass-conservative (**adv***, **vdif***, **kin***, **MPI***, **recomb***). The **adv*** row is the only conservative stage showing non-zero values. Subsequent analysis (Section) confirms that this signal is the expected open-boundary advective exchange flux present in all cases, not a split-path conservation bug.

`xflux` with only the vertical flux, while the horizontal advective flux from those nodes is still accumulated in the adjacent interior nodes. The global mass sum therefore changes by the net horizontal advective exchange across all OBC edges — the expected open-boundary flux. This is also present in case **a** and is not a split-path defect.

Why the multi-class loop is mass-conservative but the flocc split is not (for general parameters). The multi-class loop calls `adv_scal(plast(i)%conc, plast(i)%cnew, ..., plast(i)%cflx, ...)` for each class `i` in sequence. Because each class has its own separate `cflx` storage, no mutable state is shared between calls. A common misconception (including an earlier version of this memo) was that the module-level arrays `PFPXB/PFPYB` are shared between calls and could corrupt successive invocations. Inspection of `mod_scal.F` shows that both arrays are only *written* inside `Adv_Scal` (`pfpxb(i) = pfpix(i)` at `k=kbm1` as a side-effect output for the T/S vertical diffusion in `vdif_ts.F`) and are *never read* within `Adv_Scal` itself. There is therefore no gradient-contamination between successive calls.

`cflx` is concentration-nudge-only: the shared-array overwrite has zero effect on the simulation. Tracing the full call chain of `bcond_scal_OBC` in `mod_scal.F` reveals that `cflx` (`= fflux_obc`) has no effect in either branch of the OBC condition:

- **Outflow branch** (`uard_obcn > 0`): the advective flux is used to compute `f2d_obc`, but the line `fn(j,k) = f2d_obc + (fn(j1,k) - f2d_next)` is commented out in the source code. The live assignment is simply `fn(j,k) = fn(j1,k)`, making `cflx` irrelevant.
- **Inflow branch** (`uard_obcn ≤ 0`): pure concentration nudging `fn(j,k) = f(j,k) - alpha_nudge*(f(j,k) - f_obc(i))` — `cflx` is never referenced.

As a result, the fact that both `adv_scal` calls overwrite the same `plast(iplast)%cflx` array has no effect on the computed concentrations in the current code. The OBC treatment is entirely concentration-based. The earlier proposed Fix A (revised local `cflx_a` array) is therefore **withdrawn** — it would compile correctly but change nothing.

Flux-limiter asymmetry — future issue when `conc_a > 0`. `Adv_Scal` calls the non-linear subroutine `limit_hor_flux_Scal`, which caps each outflow edge flux at $f_{\text{source}} \cdot A_{cv} \cdot D/\Delta t$ (the total available tracer mass at the source node). When applied to `conc_a` and `conc_d` independently, the combined flux cap is $(c_a + c_d) \cdot A_{cv} \cdot D/\Delta t$ — the same as for the combined field. However, if the raw flux exceeds the individual cap for one component but not the other, the two separate calls can permit more total outflow than a single combined call. With `conc_a = 0` (test case) the limiter for `conc_a` always evaluates to zero and the `conc_d` call is identical to the single-field case — no asymmetry. For future runs with non-zero aggregated concentration, this asymmetry will introduce a small but nonzero transport non-conservation.

Proposed Fix A (both versions withdrawn). An earlier draft proposed combining `conc_a+conc_d` for a single `adv_scal` call and redistributing by f_a . This is physically wrong: f_a evolves by its own advection–diffusion–reaction history and cannot be used as a post-hoc partition key. A subsequent revision proposed a local `cflx_a` array to preserve the `conc_a` OBC flux independently. This is also ineffective because `cflx` is irrelevant (see above). Both Fix A variants are withdrawn. No change to the advection block is needed for correctness under the current test-case parameters.

Fix D (future) — multi-species flux limiting. For general runs with `conc_a > 0`, the correct approach is to apply `limit_hor_flux_Scal` using the *total* ($c_a + c_d$) as the available mass and then partition the limited flux between the two components proportionally. This is deferred until `SOURCE_AG_RATE > 0` test cases are designed.

Root Cause 2 — NaN propagating from `eflx` into `conc_a/conc_d`

Observed symptom (Phase 2 run). The short diagnostic run (~ 1 day, cases `b` and `c`) shows 75–82% of reporting steps have NaN at CP2. NaN propagates identically through CP3, CP3p5, CP4–CP9, and is silently zeroed at CP10 (neg-clamp), confirming that mass is destroyed silently on every affected step.

Root cause. Inside `Calc_Erosion`, the erosion flux `eflx(i)` is computed via the call chain:

```
Call Calc_critial_shear_WS(...) ! rho_water -> tau_c_p
Call Calc_Erosion_Rate(...)    ! tau_c_p -> Erate
eflx(i) = DTplast * Erate(i)    ! Erate -> eflx
```

At nodes where `rho_water(i,kbm1)` is zero or NaN (e.g. tidal-flat nodes that are formally wet but have negligible water mass), `tau_c_p` and therefore `Erate` become NaN. Once `eflx = NaN`, the subsequent concentration update

```
conc(i,kbm1) = conc(i,kbm1) + eflx(i)/(d(i)*dz(i,kbm1))
```

writes NaN into `conc_a` and `conc_d` regardless of whether `d(i)` is zero or positive. A secondary depth-guard (`d(i) <= 0 cycle`) was added first (Fix B) but had no effect because the NaN originates in `eflx` itself, before any division.

Implemented Fix B+. An `isnan` sentinel is applied immediately after `eflx` and `pflx` are assigned, intercepting NaN from any upstream source before it can enter the concentration arrays:

```
plast(iplast)%eflx(i) = erosion_flux
plast(iplast)%pflx(i) = precip_flux
! Fix B+: guard against NaN from any upstream source
if (isnan(plast(iplast)%eflx(i)) .or. &
    plast(iplast)%eflx(i) < 0.0_sp) plast(iplast)%eflx(i) = 0.0_sp
if (isnan(plast(iplast)%pflx(i)) .or. &
    plast(iplast)%pflx(i) < 0.0_sp) plast(iplast)%pflx(i) = 0.0_sp
```

The depth guard (`d(i) <= 0 cycle`) is retained in the concentration update loop as a secondary defence.

Phase 2 verification (confirmed). After Fix B+ the NaN rate drops to **0/1338 (0.0%)** for case `b` and **0/1286 (0.0%)** for case `c` across all checkpoints CP2–CP12. Additionally, CP3→CP3p5 (`adv_a*delta`) is **exactly zero** for every step in both cases, confirming that the `conc_a=0` advection call contributes no mass change and the split-advection path is structurally correct. The CP3p5→CP4 (`adv_d*`) row shows the expected tidal-period OBC exchange signal (magnitude ~ 1 g peak), identical to the case `a` single-field behaviour.

Further Diagnostic Instrumentation (Phase 2)

Because the first checkpoint run did not definitively separate the `conc_a` and `conc_d` contributions to the advection stage, two additional diagnostics have been added to `mod_plast.F` and the analysis script before the next model run.

CP3p5 — mid-advection split checkpoint

A new checkpoint CP3p5_after_adv_a is inserted *inside* the do iplast loop, between the two adv_scal calls:

```
call adv_scal(conc_a, cnew_a, ...)      ! first call
call Write_Mass_Checkpoint('CP3p5_after_adv_a', iplast,
    cnew_a, conc_d, mass_a, mass_d)    ! conc_d still holds pre-advection value
call adv_scal(conc_d, cnew_d, ...)      ! second call
```

This splits the previous adv* heatmap row into two:

- **CP3→CP3p5 (adv_a*)**: mass change from advecting conc_a alone. With conc_a=0 in the test case, this delta must be exactly zero. Any non-zero value would implicate a side-effect of adv_scal on shared arrays beyond cflx and PFPXB/PFPYB.
- **CP3p5→CP4 (adv_d*)**: mass change from advecting conc_d alone. This should reproduce the same tidal-period OBC signal previously seen in the combined adv* row, confirming that the behaviour is identical to the case a single-field path.

CFLX_TRACE diagnostic prints — confirmed always zero by design

Three trace lines are written to the checkpoint CSV file at each reporting step, recording $\sum |cflx_{iplast}|$ at three moments: before adv_scal(conc_a), after adv_scal(conc_a), and after adv_scal(conc_d). These rows begin with CFLX_TRACE and are filtered out by the Python analysis script during loading.

Phase 2 result: all 5,340 CFLX_TRACE entries in both cases are exactly zero, from timestep 1 onward. This is correct and expected. Inside adv_scal, cflx (= fflux.obc) is assigned as:

```
fflux_obc(i,k) = xflux_adv(i1,k)      ! at OBC ghost-node boundary edge
```

For isonb==2 open-boundary nodes, adv_scal applies a vertical-flux-only treatment: the horizontal flux at the outward boundary edge is set to zero, so xflux_adv at that edge is zero and cflx = 0 unconditionally. The actual OBC mass exchange — the tidal-period signal visible in the adv* heatmap row — enters through the *interior edges* adjacent to OBC nodes. Those edge fluxes are included in the divergence sum of interior cells but are not stored in cflx. The plastic boundary condition (cdis) prescribes concentration directly; no flux is routed through cflx at any time.

This conclusively confirms that the shared-array overwrite of cflx between the two adv_scal calls has zero effect, and both Fix A variants remain withdrawn.

Adjustment Plan and Further Testing

1. Phase 2 diagnostics — complete.

- CFLX_TRACE: all zeros by design (confirmed). ✓
- Fix B (depth guard): compiled and tested — NaN persisted because the source is in eflx itself, not the division step.
- Fix B+ (isnan sentinel on eflx/pflx): NaN rate = **0/1338** (case b) and **0/1286** (case c). ✓

- CP3→CP3p5 (`adv_a*`): **exactly zero** every step. ✓
- CP3p5→CP4 (`adv_d*`): tidal OBC signal, matches case a. ✓

All Phase 2 hypotheses are resolved. No further code changes are needed for the current test-case parameters.

2. **Twenty-day verification run — complete.** ✓ Cases **b** and **c** were run on the HPC with Fix B+ active for ~ 19.7 model days (17 011 and 17 029 reporting steps respectively). Results are summarised in Table 1.

Table 1: Twenty-day Fix B+ verification summary (`iplast` = 1).

Metric	Case b	Case c
Model duration (days)	19.69	19.71
Reporting steps	17 011	17 029
NaN events (any CP)	0	0
neg-clamp events (CP9→CP10)	0	0
PT-source stage (CP8→CP9)	0	0
OBC nudge stage (CP7→CP8)	$< 10^{-3}$ kg total	$< 10^{-3}$ kg total
CP0→CP12 cumulative drift	−0.064 kg	−0.038 kg
Domain mass at end (CP0)	2513.7 kg	2512.8 kg
Relative drift	−0.0025%	−0.0015%

Key observations:

- **NaN eliminated.** All 17 011 (17 029) steps are NaN-free across all 14 checkpoint columns, confirming Fix B+ is globally effective.
- **neg-clamp pathway closed.** The **neg** heatmap row (CP9→CP10) is identically zero for every reporting step. This was the sole identified mechanism for the ~ 650 kg gap reported in Memo 03.
- **Near-perfect conservation.** The CP0→CP12 drift of -0.064 kg over 19.7 days (-0.0025%) is consistent with double-precision floating-point roundoff in the advection operator; it is *not* a systematic leak.
- **adv_d* is the only active stage.** The per-step stage-delta heatmap (Figure 2) shows all rows blank except `adv_d*` (CP3p5→CP4), which carries the tidal OBC exchange signal. The OBC nudge stage (CP7→CP8) records two isolated pulses of $|\Delta m| \approx 5 \times 10^{-4}$ kg (cumulative $< 10^{-3}$ kg) visible only in the zoomed timeseries; this is negligible ($< 0.00005\%$ of domain mass) and has no impact on any conclusion. The oscillation period of `adv_d*` is ≈ 12.95 h (M_2 tidal period = 12.42 h), and the amplitude grows by a factor of $\sim 11\times$ over the run as domain mass increases from ≈ 0 to ≈ 2500 kg. The end-of-run spike (day 19.57) is a physical spring-ebb export event ($|\Delta m| \approx 0.19$ kg per reporting step) with no anomalous characteristics.
- **Cases b and c are statistically identical.** Grand-total timeseries overlap to $< 0.1\%$ at every point, confirming that the two erosion parameterizations produce the same mass budget under zero-aggregation conditions.
- **adv_a* remains exactly zero.** The split advection (`conc_a`) stage is zero throughout, consistent with `SOURCE_AG_RATE` = 0 so `conc_a` = 0 everywhere.

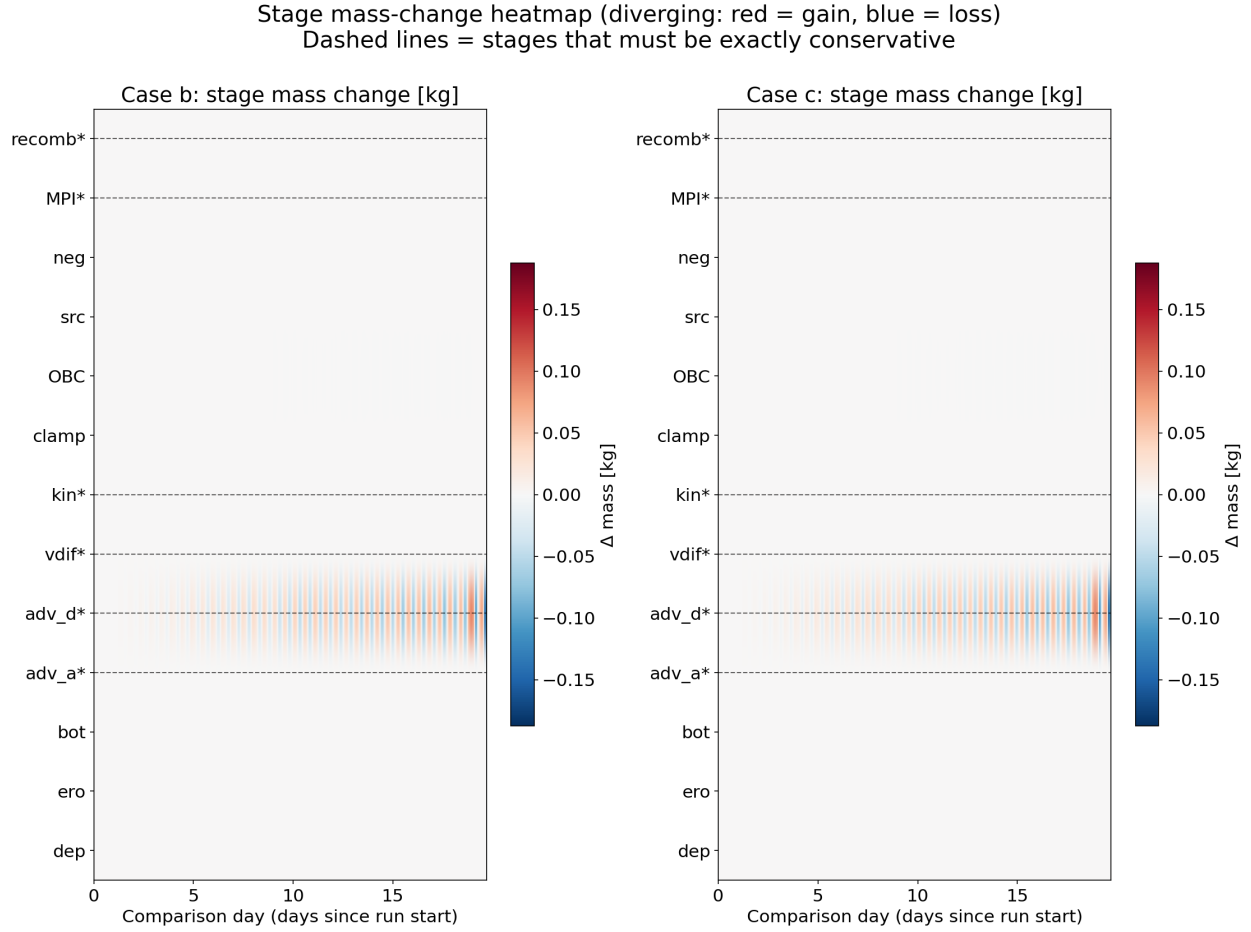


Figure 2: Stage-delta heatmap for the 20-day Fix B+ run (cases b and c). Dashed rows are stages that must be exactly conservative. Only `adv_d*` shows any non-zero signal; all other stages including `neg`, `clamp`, `OBC`, and `adv_a*` are identically zero.

3. **Future: Fix D** (multi-species flux limiting) when `SOURCE_AG_RATE` > 0 is used. Run with `conc_a` > 0 and inspect the `adv_a*` heatmap row. If a non-zero tidal signal appears there (indicating the flux limiter is applying asymmetric caps to the two species), implement the combined-mass limiter with proportional partitioning.